



Operating Instructions

Diesel engine-generator set

mtu 8V1600 DS400 / DG08V1600A1E

mtu 8V1600 DS440 / DG08V1600A2E

mtu 10V1600 DS500 / DG10V1600A1E

mtu 10V1600 DS560 / DG10V1600A2E

mtu 12V1600 DS660 / DG12V1600A1E

mtu 12V1600 DS730 / DG12V1600A2E

Controller DSE8610, DSE8620, DSE8660

SS180027/05E

System designation	Engine model	Application group *
mtu 8V1600 DS400 / DG08V1600A1E	8V1600G10F	3B
mtu 8V1600 DS400 / DG08V1600A1E	8V1600G70F	3D
mtu 8V1600 DS440 / DG08V1600A2E	8V1600G20F	3B
mtu 8V1600 DS440 / DG08V1600A2E	8V1600G80F	3D
mtu 10V1600 DS500 / DG10V1600A1E	10V1600G10F	3B
mtu 10V1600 DS500 / DG10V1600A1E	10V1600G70F	3D
mtu 10V1600 DS560 / DG10V1600A2E	10V1600G20F	3B
mtu 10V1600 DS560 / DG10V1600A2E	10V1600G80F	3D
mtu 12V1600 DS660 / DG12V1600A1E	12V1600G10F	3B
mtu 12V1600 DS660 / DG12V1600A1E	12V1600G70F	3D
mtu 12V1600 DS730 / DG12V1600A2E	12V1600G20F	3B
mtu 12V1600 DS730 / DG12V1600A2E	12V1600G80F	3D
* 3B Continuous service, variable load, ICXN (PRP) 3D 3D, emergency standby power, IFN (ESP)		

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

© Copyright Rolls-Royce Solutions

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of Rolls-Royce Solutions. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage and/or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

Table of Contents

1	Preface		
1.1	Preface	6	
2	Safety		
2.1	Important provisions for all products	7	
2.2	Intended use of all products	8	
2.3	Personnel and organizational requirements	9	
2.4	Safety regulations for commissioning and operation	10	
2.5	Safety regulations for assembly, maintenance, and repair work	12	
2.6	Fire and environmental protection, indirect materials	16	
2.7	Standards for warning messages in the text and highlighted information	19	
2.8	Overview of safety signs on engine-generator set	20	
2.9	Protective devices on engine-generator set - Overview	22	
3	Transport		
3.1	Transport	24	
4	Product Summary		
4.1	Engine-generator set - General information	25	
4.2	Engine-generator set - Overview	27	
4.3	Engine-generator set - Control versions	29	
4.4	Engine	30	
4.4.1	Engine side and cylinder designations	30	
4.4.2	Engine - Overview	31	
4.4.3	Sensors - Overview	32	
5	Technical Data		
5.1	Technical data mtu 8V1600 DS400, application group 3B (PRP)	34	
5.2	Technical data mtu 8V1600 DS440, application group 3B (PRP)	37	
5.3	Technical data mtu 10V1600 DS500, application group 3B (PRP)	39	
5.4	Technical data mtu 10V1600 DS560, application group 3B (PRP)	41	
5.5	Technical data mtu 12V1600 DS660, application group 3B (PRP)	43	
5.6	Technical data mtu 12V1600 DS730, application group 3B (PRP)	45	
5.7	Technical data mtu 8V1600 DS400, application group 3D (ESP)	47	
5.8	Technical data mtu 8V1600 DS440, application group 3D (ESP)	49	
5.9	Technical data mtu 10V1600 DS500, application group 3D (ESP)	51	
5.10	Technical data mtu 10V1600 DS560, application group 3D (ESP)	53	
5.11	Technical data mtu 12V1600 DS660, application group 3D (ESP)	55	
5.12	Technical data mtu 12V1600 DS730, application group 3D (ESP)	57	
5.13	Firing order	59	
6	Operation		
6.1	Preparations for putting into operation after extended out-of-service periods	60	
6.2	Engine-generator set - Start	63	
6.3	Re-starting the engine following an automatic safety shutdown	64	
6.4	Operational monitoring	65	
6.5	Engine-generator set - Shutdown	66	
6.6	Emergency shutdown	67	
6.7	After shutdown - Putting engine-generator set out of operation	68	
7	Maintenance Schedule		
7.1	Preface	69	
7.2	Maintenance task tolerances	70	
7.3	Maintenance schedule matrix	71	
7.4	Maintenance tasks	72	
8	Troubleshooting		
8.1	Troubleshooting	73	
8.2	Fault messages	75	
9	Task Description		
9.1	Engine-Generator Set	90	
9.1.1	Engine-generator set - Test run	90	
9.2	Engine	91	
9.2.1	Engine - Barring with starting system	91	
9.3	Valve Drive	92	
9.3.1	Valve clearance - Check and adjustment	92	
9.3.2	Cylinder head cover - Removal and installation	95	
9.4	Fuel System	98	

9.4.1 Fuel system - Venting	98	10.3 Operating Manual DSE8660	269
9.5 Fuel Filter	99	10.4 Quick Start Guide DSE8620	349
9.5.1 Fuel filter - Replacement	99	10.5 Operating Manual DSE8620	389
9.5.2 Fuel prefilter - Draining	100		
9.5.3 Fuel prefilter - Replacement	101	11 Fluids and Lubricants Specifications	
9.6 Air Filter	102	11.1 Preface	511
9.6.1 Air filter - Replacement	102	11.1.1 General information	511
9.6.2 Air filter - Removal and installation	103	11.2 Engine Oils	513
9.7 Air Intake	104	11.2.1 Requirements and oil change intervals	513
9.7.1 Service indicator - Signal ring position check	104	11.2.2 Viscosity grades	515
9.8 Starting Equipment	105	11.2.3 Series-dependent usability of engine oils	516
9.8.1 Starter - Condition check	105	11.2.4 Used-oil analysis	517
9.9 Lube Oil System, Lube Oil Circuit	106	11.2.5 mtu Advanced Fluid Management System for engine oils - Test package for North America	519
9.9.1 Engine oil - Level check	106	11.3 Coolants	520
9.9.2 Engine oil - Change	107	11.3.1 Coolant - General	520
9.10 Oil Filtration / Cooling	109	11.3.2 Unsuitable materials in the coolant circuit	522
9.10.1 Engine oil filter - Replacement	109	11.3.3 Requirements imposed on freshwater	523
9.11 Coolant Circuit, General, High-Temperature Circuit	110	11.3.4 Operational checks	524
9.11.1 Engine coolant - Level check	110	11.3.5 Storage capability of coolant concentrates	525
9.11.2 Engine coolant - Change	111	11.3.6 Color additives to detect leakage in the coolant circuit	526
9.11.3 Engine coolant - Draining	112	11.3.7 Series-dependent usability of coolant additives	527
9.11.4 Engine coolant - Filling	113	11.3.8 mtu Advanced Fluid Management System for coolant - Test package for North America	528
9.11.5 Engine coolant pump - Relief bore check	115	11.4 Liquid Fuels	530
9.11.6 Coolant cooler - External contamination and leak-tightness check	116	11.4.1 Diesel fuels - General information	530
9.11.7 Coolant cooler - Cleaning	117	11.4.2 Biodiesel - biodiesel admixture	534
9.11.8 Preheater - Coolant line check	119	11.4.3 Heating oil EL	535
9.12 Belt Drive	120	11.4.4 Supplementary fuel additives	536
9.12.1 Drive belt - Adjustment	120	11.4.5 Series-dependent approval of diesel fuel grades for mtu engines	538
9.12.2 Drive belt - Condition check	121	11.4.6 Unsuitable materials in the diesel fuel circuit	543
9.12.3 Drive belt - Tension check	122	11.4.7 Measures prior to engine out-of-service periods >1 month	544
9.12.4 Drive belt - Replacement	124	11.4.8 mtu Advanced Fluid Management System for fuels - Test package for North America	545
9.13 Wiring (General) for Engine/Gearbox/Unit	126	11.5 NOx Reducing Agent AUS 32 / AUS 40 for SCR Exhaust Gas Aftertreatment Systems	547
9.13.1 Engine cabling - Check	126	11.5.1 General information and storage	547
9.14 Accessories for (Electronic) Engine Governor / Control System	127	11.6 Approved Engine Oils and Lubricating Greases	548
9.14.1 Engine governor and connectors - Cleaning	127	11.6.1 Multi-grade oils - Category 2 of SAE grades 10W-40, 15W-40 and 20W-40 for diesel engines	548
9.14.2 Engine governor - Checking plug connections	128	11.6.2 Multi-grade oils - Category 2.1 (Low SAPS oils), SAE grades 0W-30, 10W-30, 5W-40, 10W-40 and 15W-40	557
9.15 Generator	129	11.6.3 Multi-grade oils - Category 3, SAE grades 5W-30, 5W-40, 10W-40 and 15W-40 for diesel engines	561
9.15.1 Generator - Check	129		
9.15.2 Generator - Wiring check	130		
9.15.3 Generator mounting - Securing screw firm seating check	131		
9.15.4 Generator - Regreasing antifriction bearing	132		
10 Monitoring and Control			
10.1 Quick Start Guide DSE8610	135		
10.2 Operating Manual DSE8610	159		

11.6.4	Multi-grade oils - Category 3.1 (Low SAPS oils), SAE grades 5W-30, 10W-30 and 10W-40 for diesel engines	566	11.8.5	Cleaning engine coolant circuit assemblies	582
11.6.5	Lubricating greases for diesel engine-generator set components	572	11.8.6	Coolant circuits contaminated with bacteria, fungi or yeast	583
11.7	Approved Coolants	573	11.9	Cleaning	584
11.7.1	Antifreeze concentrates on ethylene glycol basis	573	11.9.1	General information	584
11.7.2	Antifreeze ready mixtures on ethylene glycol basis	576	11.9.2	Approved cleaning agents	585
11.8	Flushing and Cleaning Specifications for Engine Coolant Circuits	578	12	Appendix A	
11.8.1	General information	578	12.1	Abbreviations	586
11.8.2	Approved cleaning agents	579	12.2	Contact person/Service partner	589
11.8.3	Engine coolant circuits - Flushing	580	13	Appendix B	
11.8.4	Engine coolant circuits - Cleaning	581	13.1	Standard Tools and Special Tools	590
			13.2	Index	594

1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

The operating instructions are divided into the following chapters:

- Safety
- Transport
- Product overview
- Technical data
- Operation
- Maintenance Schedule
- Troubleshooting
- Task description
- Monitoring and control
- Fluids and Lubricants Specifications

Store these operating instructions where they can be accessed by all persons entrusted with operating or servicing the product.

Read these Operating Instructions carefully before starting any work. For safe operation of the product, always follow all safety instructions and work instructions.

Local accident prevention regulations and general safety regulations for the area of application of the product also apply.

These Operating Instructions form part of the product and refer to the latest design version of the product at the time of publication. They are subject to modification to reflect technical enhancements. Actual product may vary from figures shown.

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at: <http://www.mtu-solutions.com>

2.2 Intended use of all products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

The product must not be operated in explosive atmospheres unless the engine fulfills the conditions for such use and approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

2.4 Safety regulations for commissioning and operation

Safety regulations for commissioning

Install the product correctly and carry out acceptance in accordance with the manufacturer's specifications before commissioning the product. All necessary approvals must be granted by the relevant authorities and all requirements for commissioning must be fulfilled.

Whenever the product is subsequently put into operation ensure that:

- All personnel is clear of the danger zone surrounding moving parts of the machine.
Electrically-actuated linkages may be set in motion when the Engine Control Unit (governor) is switched on.
- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- All protective devices are in place.
- All lines bearing media are ready for operation.
- The walkways in the operating room are unobstructed and safe.
- No persons wearing pacemakers or any other technical body aids are present.
- Adequate ventilation of the operating room is ensured for any operating status.
- In the first few hours of operation, the product emits gases as a result of smoldering e.g. lacquers or oil.
These gases may be hazardous to health. Always wear respiratory protection in the operating room during this period.
- The exhaust system is leak-tight and the gases are vented to atmosphere.
- The product is free of any damage, this applies in particular to lines and cabling.
- All components are properly grounded. Ground separately by means of a grounding stake as necessary.
- The power supply (mains plug) is always freely accessible to facilitate disconnecting the device or system from the power supply at any time.
- Battery terminals, generator terminals or cables are protected against accidental contact.
- All connections have been correctly allocated e.g. +/- polarity, fuel line/reducing agent line, supply/return.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems work properly.

Smoking is prohibited in the area of the product.

Safety regulations during operation

The operator must be familiar with the control and display elements.

The operator must be familiar with the consequences of any operations performed.

During operation, the display instruments and monitoring units must be observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

Practice emergency procedures, especially emergency stopping, at regular intervals.

Take the following steps if any system malfunctions are detected or signaled by the system:

- Inform supervisor(s) in charge.
- Analyze the message.
- Respond by taking any necessary emergency action, e.g. emergency stop.

After a safety shutdown, the engine may only be started after the cause of the shutdown has been eliminated.

Contact Service if the root cause of the malfunction cannot be clearly identified.

Operation

Do not remain in the operating room when the product is running unless absolutely necessary. Keep your stay as short as possible.